A Better Alarm

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Project Summary

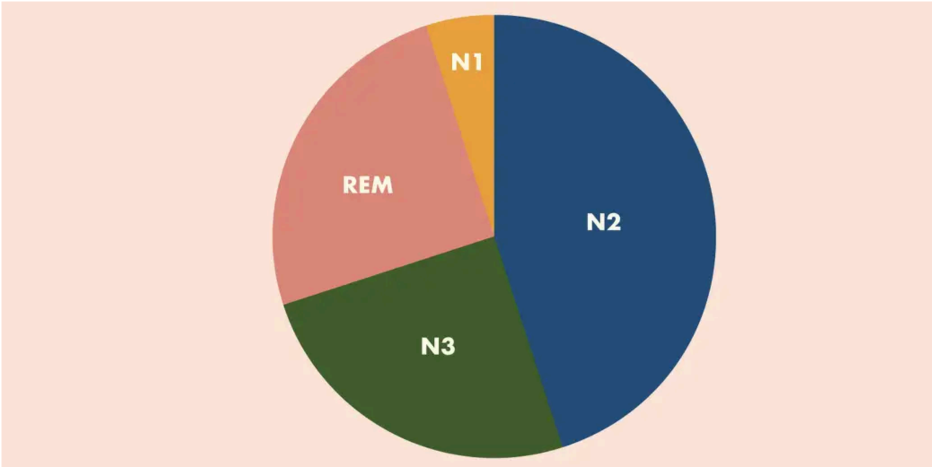
In this project, we propose a solution for individuals at risk of developing health issues due to irregular waking up. We have conducted surveys to determine whom this problem may directly affect and work with experts to find an effective solution. Health risks may include a heart attack, a spike in blood pressure, or a stroke by waking up abruptly. With innovative technology, we will develop a solution that can help reduce the potential health risks of a person who is suddenly awakened.

Problem Statement

80% of teens and adults get awakened abruptly many mornings, causing health risks. According to the National Center on Sleep Disorders Research, the act of waking abruptly activates your flight or fight reaction, which in turn increases blood pressure, stress, and can cause other cardiovascular problems.

Statistics

“Kim compared morning blood pressure surge measures between the natural and forced awakening scenarios. Although the results from this pilot must be interpreted with caution and validated in a larger sample, her research showed that those who were forced awake had a morning blood pressure surge that was 74% greater than those who awoke naturally – evidence of a link between short sleep duration, forced awakening and morning blood pressure surge. Adults with cardiovascular disease might experience more of the adverse effects of a morning blood pressure surge when they get little sleep and are jolted awake” (Berry, A. (2023, December 19). *Waking Up to Your Phone Alarm? It Could Be Putting You at Risk.*



Sleep Cycle:

Sleep stage	**Alternative name**	**Type of sleep**	**% of total sleep**
Stage 1	N1	Light sleep/NREM	1-5%
Stage 2	N2	Light sleep/NREM	45-50%
Stage 3	N3	Deep sleep/NREM	20-25%
Stage 4	REM	Rem sleep/REM	20-25%

Survey Link

https://forms.gle/BkUsGTuTe7LwVDFK6

Existing Solutions

There have been several attempts to address the issue of waking up suddenly, with various solutions and patents, including natural light alarm clocks, shock watches, and sleep phones. While these solutions are viable, each has its flaws. The natural light alarm clock is a popular option, but it may not be effective for deep sleepers, sometimes failing to wake the user and defeating the purpose of an alarm clock. Other alternatives, such as shock alarm clocks and those that require shaking to turn off, tackle similar issues but do not appear to be the best solution due to multiple issues.

Sleep Experts

Sleep experts like Antony Terence believe that waking up suddenly can lead to a condition known as “sleep inertia." Research from the UVA School of Nursing indicates that waking up abruptly to a loud noise, such as an alarm clock, may raise your blood pressure, increasing your risk for adverse cardiovascular events, including stroke and heart attack.

How the information was found

Our information was acquired by reading articles from multiple credible sources, such as News.virginia.edu., ABC News, and ADCO Hearing Products. After looking through prior solutions, we have found that the best way to wake up naturally is with a light that gradually wakes you up, similar to the sun waking you up in the morning. However, a sound-based alarm may be one of the worst forced awakening scenarios as it has been proven to raise your blood pressure 74% more than a person who naturally wakes up. Experts such as Yeonsu Kim, a nursing doctoral student, and experts at the National Institute of Industrial Health in Japan have conducted studies to support their arguments against the use of devices that would abruptly wake you up.

List of Patents

1. Loftie

U.S Patent: USD923490

Date of Patent: June 21, 2021

Inventor: Gregory John Wolos

Strengths: Includes a variety of sounds, playlists, stories, and meditations; features a soft light

Shortcomings: Expensive; Sounds must be selected manually, and buttons take getting used to

2. Philips SmartSleep Wake-up Light

U.S Patent: USD690455S1

Date of Patent: September 24, 2013

Inventor: Georg Johann Hagenauer

Strengths: Gently wakes you up; Has a setting that can help you sleep better

Shortcomings: The price is too high. If you are a heavy sleeper, it may not be effective as the light may not be bright enough

3. Fitbit Versa

U.S Patent: 11877861

Date of Patent: January 23, 2024

Inventor: Conor Joseph Heneghan, Jacob Anthony Arnold, Zachary Todd Beattie, Alexandros A. Pantelopoulos, Allison Maya Russell, Philip Foeckler, Adrienne M. Tucker, Delisa Lopez, Belen Lafon, Atiyeh Ghoreyshi

Strengths: Tracks sleep patterns,

Shortcomings: Smartwatch productivity features are lacking

4. AcousticSheep SleepPhones

U.S Patent: US8213670B2

Date of Patent: 2012-07-03

Inventor: Wei-Shin Lai

Strengths: 12hr battery life, Good noise isolation, effective in all sleeping positions

Shortcomings: No active noise cancellation; Pricey for the headband style

5. Shock Clock 3

U.S Patent: 202421775U
Date of Patent: 2012-09-05
Inventor: 赵紫玉
Strengths: lightweight and comfortable, sleep tracking
Shortcomings: Vibrations, even at full power, are very weak, so they may not wake you up

All the products are proven to be reasonable solutions, though not perfect. Each has specific niches that need improvement, and we identified the issues by reading reviews from both the product's website and third-party sellers. We considered multiple reviews from individuals who have purchased that specific product. We should trust these reviews because the buyers have good intentions and no incentive to mislead us, as they aren't sponsored or compensated by the product.

Design Requirements

1. Safety, Legal, and Ethical Issues - **Safety issues could be heat problems with suffocation from a pillow, possibly causing a fire hazard.**
2. Performance - **The product must be able to wake up even the heaviest sleepers regularly while being gentle enough not to harm light sleepers**.
3. Customer needs - **The customer wants a device powerful enough to consistently wake up early and not sleep in while not being startled or harmed during the process.**
4. Size and Weight - **Restrictions to both as the product needs to be smaller than 20" by 26 inches. and high-powered while lightweight to still remain comfortable.**
5. Materials - **Circuits and a motherboard for electrical power. Fabric or polyester to be soft and comfortable.**
6. Target Cost - **Around $20-$25**
7. Durability and Maintenance - **Will require working batteries to be maintained.**
8. Operating Environment - **Needs to be manufactured and able to vibrate and ventilate to reduce the risk of overheating and possible fire hazards (must be ventilated even through a pillow).**
9. Ergonomics - **Human comfort, such as sleeping on a rigid, hard surface, is neither comfortable nor ideal.**
10. Product Life - **5-10 years due to a mechanical breakdown and loss of efficiency.**
11. Aesthetics - **Preferences include a product that is not rigid and flexible to remain comfortable under one's pillow.**
12. Environmental Impact and Sustainability - **Will not include toxic or dangerous substances and should not be disposed of; just replace batteries.**

Our product cannot drastically increase heart rate. Additionally, it must effectively wake deep sleepers while ensuring consumer safety. It should be comfortable and awaken our users gently without being abrupt. We met as a team and reviewed all our notes throughout the project to prioritize design specifications in a list from 1 to 12, ultimately targeting the world's working class. We will determine whether our product will be successful when we test the design ourselves. Through our survey, we have gathered valuable information about our target market.

Citations

Berry, A. (2023, December 19). *Waking Up to Your Phone Alarm? It Could Be Putting You at Risk*. News.virginia.edu. https://news.virginia.edu/content/waking-your-phone-alarm-it-could-be-putting-you-risk

Ramsey, L. (2023, August 30). *Does Your Alarm Sound Affect Your Mood?* News-Medical.net. https://www.news-medical.net/health/Does-Your-Alarm-Sound-Affect-Your-Mood.aspx

USPTO. (2024). *United States Patent and Trademark Office*. Uspto.gov. https://www.uspto.gov/